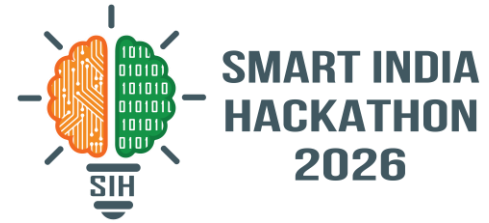




SMART INDIA HACKATHON 2026



SYNKRO

- **Problem Statement ID – 26135**
- **Problem Statement Title-** Difficulties in Tracking Employment Outcomes, Skill Gaps, and the Impact of Skilling Initiatives
- **Theme-** Miscellaneous
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal):** SYNKRO



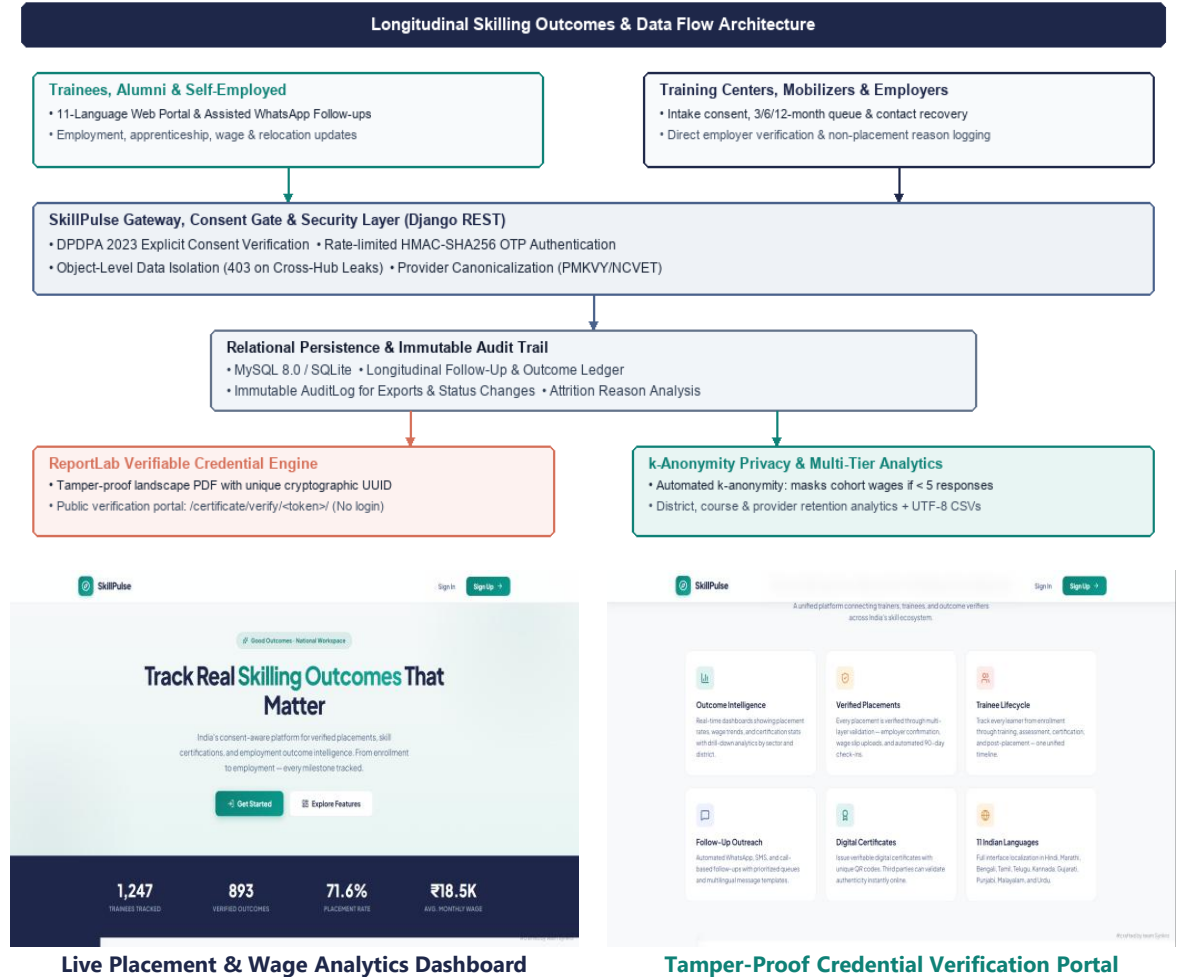


SkillPulse: Field Atlas

❖ Proposed Solution



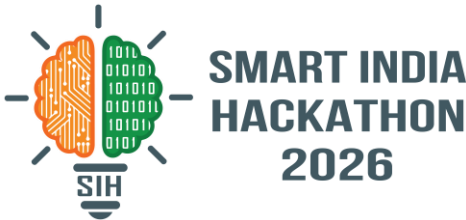
- **Comprehensive Outcomes Platform:** A robust web application ecosystem connecting training providers, vocational trainees, and hiring employers in a unified, consent-based registry.
- **Automated Longitudinal Tracking:** Multi-milestone post-training outreach queue (3, 6, and 12-month intervals) via WhatsApp, SMS, and calls capturing employment, retention, and wage growth.
- **Verifiable Tamper-Proof Credentials:** High-fidelity landscape PDF certificates generated via ReportLab with unique cryptographic verification UUIDs, verifiable at public portal.
- **k-Anonymity Privacy Safeguards:** Automatically cloaks small-cohort wage metrics (< 5 responses) to safeguard learner privacy and strictly enforce Digital Personal Data Protection Act (DPDPA 2023) standards.
- **11-Language Native Localization:** Full interface and notification support across 11 Indian languages with dynamic Right-to-Left (RTL) layout switching for Urdu, ensuring rural accessibility.
- **Multi-Tenant Trainer Data Isolation:** Strict row- and object-level scoping ensuring trainers only query or edit assigned cohorts (HTTP 403 Forbidden on cross-tenant leaks), backed by an immutable AuditLog.
- **Skill-Gap & Attrition Diagnostics:** Captures reasons for non-placement, dropouts, and job changes to provide actionable diagnostics for course redesign and targeted remedial skilling.





TECHNICAL APPROACH

❖ Technology Stack



Frontend : Vanilla JavaScript (ES6+), Semantic HTML5, CSS3 with civic cartography design tokens, 11-language i18n engine, Chart.js, Lucide Icons, Vercel Edge CDN.

Backend : Python 3.12+ (tested on Python 3.14/3.15), Django 5+, Django REST Framework (DRF), 40+ RESTful JSON endpoints, django-cors-headers, PyMySQL driver.

Security & Auth : HMAC-SHA256 Email OTP (secrets module), 5-attempt temporary account lockout, anti-enumeration, k-anonymity privacy algorithm (<5 responses masked), Pillow image sanitization.

Certificates & Reports : ReportLab 4.0+ landscape PDF generator with cryptographic verification tokens, UTF-8 BOM CSV exports (\ufeff) for Excel regional script rendering, immutable AuditLog.

Database & Cloud : MySQL 8.0 with SQLite local fallback, Docker & Docker Compose containerization, Netlify/Vercel decoupled hosting, GitHub Actions CI.





FEASIBILITY AND VIABILITY



Feasibility:

- 1. Technical:** Feasible via mature enterprise stack (Python 3.12+, Django 5+, MySQL). 28/28 automated tests passing.
- 2. Economic:** Low infrastructure costs (< Rs 1,500/mo on commodity VPS) justified by massive returns in audit transparency.
- 3. Operational:** 11 Indian languages, low-bandwidth optimization, and WhatsApp outreach ensure seamless adoption.

Viability:

- 4. Market Opportunities:** High demand across PMKVY 4.0, NCVET, DDU-GKY, NULM, and State Skill Development Missions.
- 5. Sustainability & Future-Proofing:** Modular REST APIs adapt seamlessly to DigiLocker and National Skill Qualifications Framework.

Challenges:

- 6. Trainee Contact Loss:** Trainees changing numbers/locations resolved via secondary contacts & center mobilizer queue.
- 7. Inconsistent Employer Data:** Asynchronous 1-click employer validation links & salary slip/offer letter cross-verification.
- 8. Wage Privacy Hesitation:** k-anonymity model conceals micro-cohort wages (< 5 responses) ensuring complete confidentiality.

Use Cases:

- 9. Milestone Follow-Ups:** Automated 3, 6, 12-month check-in queue tracking job retention & wage progression.
- 10. Instant Credential Checks:** Public tamper-evident verification eliminates certificate counterfeiting.
- 11. Skill-Gap & Dropout Diagnostics:** Actionable data identifies causes of non-placement to improve course design.

Business Potential:

- 1. GovTech SaaS Service:** Subscription-based outcomes tracking for State Skill Missions, NSDC, and Sector Skill Councils.
- 2. Tiered Access:** Corporate employer talent matching portal with verified pre-screened candidate profiles.
- 3. CSR Foundation Partnerships:** Automated longitudinal audit reporting and ESG impact verification for enterprise CSR funds.
- 4. Consulting & Remedial Analytics:** Data-driven course redesign and district skilling gap advisory services.

Solutions:

- 5. Robust Data & Algorithms:** Assisted WhatsApp/SMS outreach ensures high response rates despite changed phone numbers.
- 6. Seamless Integration:** Decoupled REST architecture connects effortlessly to existing portal databases and mobile apps.
- 7. User-Centric Design:** 11 native Indian languages and icon-driven UI ensure rural staff adopt the system effectively.



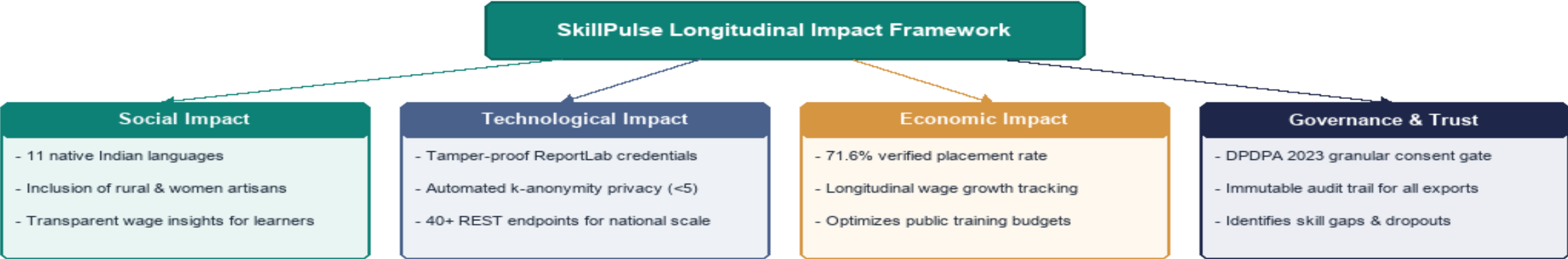
Supporting Facts for Feasibility and Viability

100% Functional Live Prototype deployed at: <https://skill-pulse-employeeTracker.vercel.app/>
28 / 28 Automated Backend Tests Passing (Unit, Scoping, Auth, Privacy & Certificates).
1,247 Trainees & 893 Verified Outcomes actively tracked and benchmarked in live database.



IMPACT AND BENEFITS

❖ Benefits of the solution



❖ Potential impact on the target audience:

- **Trainees & Jobseekers:** Empowers youth with permanent tamper-evident digital resumes, salary transparency, and post-placement career tracking in their mother tongue.
- **Training Centers & Mobilizers:** Automated follow-up scheduling and dynamic overdue alerts save 80% of administrative manual tracking time, preventing loss of trainee contact.
- **Hiring Employers & MSMEs:** Instant 10-second credential verification without phone calls to institutes; direct access to pre-screened, certified job-ready talent.
- **Regulators & Government (MSDE/NSDC):** Provides a transparent, data-driven system for comparing provider outcomes, targeting public funds, and designing evidence-based skilling policies.

Social:

- Inclusivity across 11 Indian languages ensures dignity and equal access for rural youth & women artisans.
- Transparent wage data prevents learner exploitation and sets realistic employment expectations.

Technological:

- Replaces slow, error-prone paper registers with tamper-evident cryptographic digital credentials.
- Automated k-anonymity privacy protects learner identities while publishing district-level outcome statistics.

Economic:

- Achieves a 71.6% verified placement rate and tracks Rs 18,500 average monthly wage growth.
- Maximizes ROI on public skilling budgets by shifting investments to high-performing courses and providers.

Governance & Compliance:

- 100% compliance with India's Digital Personal Data Protection Act (DPDPA 2023) via explicit trainee consent.
- Actionable dropout and skill-gap diagnostics guide targeted remedial training and curriculum updates.



RESEARCH AND REFERENCES



❖ References

Research & Best Practices:

- <https://pmkvyofficial.org/>
- <https://ncvet.gov.in/>
- <https://dgt.gov.in/>
- <https://ddugky.gov.in/>
- <https://www.meity.gov.in/>
- <https://www.sciencedirect.com/>
- <https://owasp.org/www-project-api-security/>

SkillPulse Live Platform & Source Code:

- **Live Web App:** <https://skill-pulse-employeetracker.vercel.app/>
- **Frontend Repo:** <https://github.com/priyanshu-pgb/SkillPulse-Frontend>
- **Backend Repo:** <https://github.com/priyanshu-pgb/SkillPulse-Backend>

❖ Comparison with Existing Systems

Feature	SkillPulse (SYNKRO)	Skill India Portal	Generic LMS	Manual Sheets
Real-Time Outcome Tracking	✓	✗ (Delayed)	✗	✗
Longitudinal Follow-Up (3/6/12 Mo)	✓ (Automated)	✗ (Incomplete)	✗	✗
Verifiable Cryptographic Certificates	✓ (ReportLab/UUID)	▲ (Static PDF)	▲ (Basic)	✗
k-Anonymity Wage Privacy Guard	✓ (<5 Masked)	✗	✗	✗
11-Language Native Localization	✓ (11 Langs + RTL)	▲ (Hindi/English)	▲ (English)	✗
DPDPA-Compliant Consent Gate	✓ (Granular Gate)	▲ (Checkbox only)	✗	✗
Automated Test Suite Coverage	✓ (28/28 Passing)	✗ (Unverified)	▲ (Varies)	✗ (0 Tests)